

ILIAD Intensive

# Computational Mechanics

## 1. Overview & Scope

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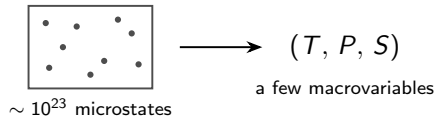
Autumn 2026

# Outline

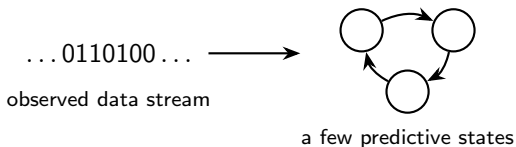
1. What is Computational Mechanics?
2. Computational Mechanics & AI Safety
3. Overview: where is the program at?
4. Scope: what will we tackle today?

# What is Computational Mechanics?

**Statistical Mechanics** makes predictions about the behaviour of systems by considering their **statistical** properties:



**Computational Mechanics** makes predictions about the behaviour of systems by considering their **computational** properties:



# Central questions

Some questions that are central to comp-mech:

- How much of the past does a system store?
- How does the system store this information?
- How does this information affect system behaviour?

We will consider each of these questions for transformers today!

# Optimal prediction

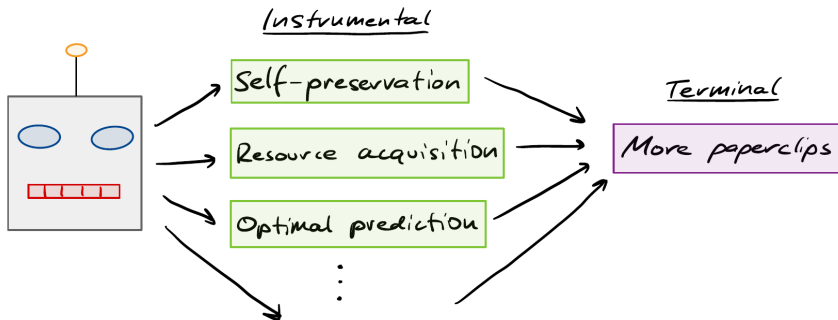
Not all systems are interesting! Comp-mech places an emphasis on those performing **optimal prediction**. In this setting the central question becomes:

What are convergent structures relevant for **optimal prediction**?

Insight on this question greatly constrains our search space when considering **optimal predictors**.

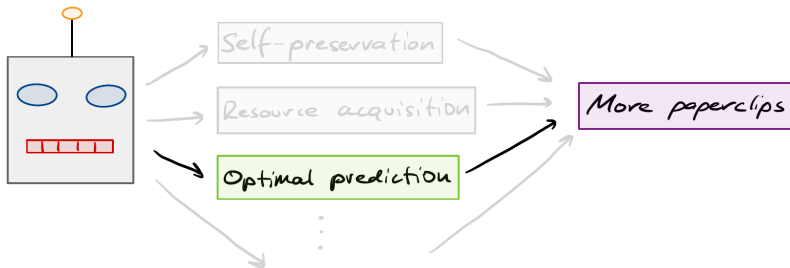
# Instrumental and terminal goals

We have strong reason to believe that AI systems will pursue a range of **instrumental goals** in service of their **terminal goal**.



# Optimal prediction as an intervention target

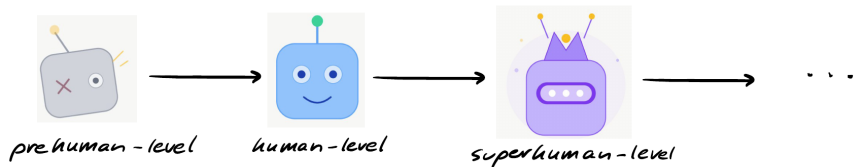
Comp-mech uses an **instrumental goal** as an intervention target point.



By considering internal representations consistent with optimal prediction the search space is reduced.

# Capability scale and optimal prediction

Targeting an instrumental goal is **scale-free**: it will be relevant to current and future models.



Which contrasts **scale-sensitive** approaches e.g. AI control that are only applicable up to a fixed capability scale.

Moreover, as capabilities increase models will better approximate **optimal predictors**.

# The research program

Comp-mech applied to AI safety is a young research program ( $\sim 2.5$  years).  
There is much to do!

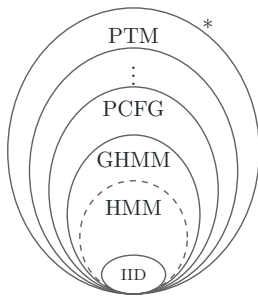
Progress can be broken down into two orthogonal directions:

- For what classes of data generating **processes** do we understand optimal prediction?
- For what scale of **neural networks** have we identified the structures of optimal prediction?

# Processes and generation resources

Fix a (computable) stochastic process and consider:

What resources are required to generate this process? This naturally leads to a hierarchy:

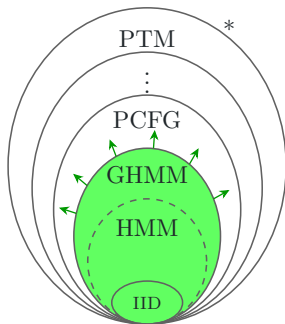


\* not exhaustive

|      |                                       |
|------|---------------------------------------|
| PTM  | Probabilistic Turing Machine          |
| PCFG | Probabilistic Context-Free Grammar    |
| GHMM | Generalised Hidden Markov Model       |
| HMM  | Hidden Markov Model                   |
| IID  | Independent & Identically Distributed |

# Processes and optimal prediction

For what classes of data generating **processes** do we understand optimal prediction?



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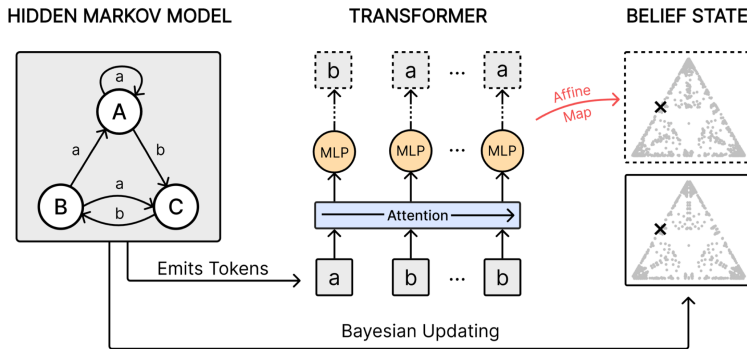
# Neural networks: the scale studied so far

For what scale of **neural networks** have we identified the structures of optimal prediction?

| Architecture | Parameters (approximately) |
|--------------|----------------------------|
| Transformer  | $10^5$                     |
| LSTM         | $10^5$                     |
| RNN          | $10^4$                     |
| GRU          | $10^5$                     |

Very toy compared to current frontier models ( $\sim 10^{12}$ )!

# Example: belief state geometry

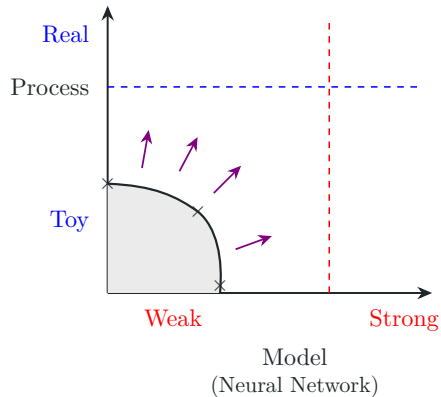


After next-token training on HMM sequences, an affine readout approximately recovers the geometry of beliefs over hidden states from residual activations.

Shai et al., *Transformers represent belief state geometry in their residual stream*. NeurIPS 2024. [arXiv:2405.15943](https://arxiv.org/abs/2405.15943).

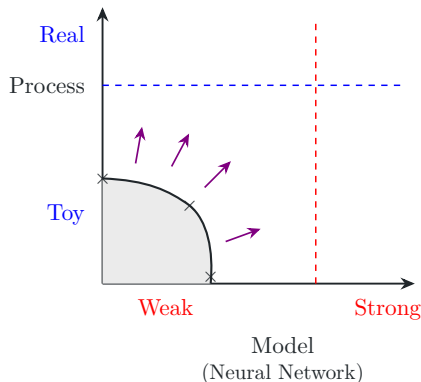
# The research frontier

We are working to push the Pareto frontier:



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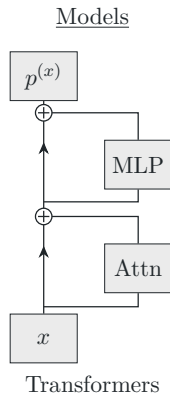
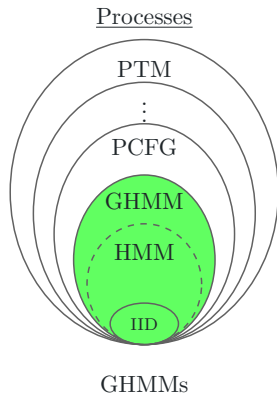


## Wild Speculation

As the research program is scale-free one view is that we should develop the foundations such that future automated AI Safety researchers can pursue these ideas independently.

# Scope: what will we tackle today?

We will take a slightly narrower view and consider:



# Plan

- |                                   |                    |
|-----------------------------------|--------------------|
| 1 Predicting the future           | Lecture/Exercises  |
| 2 Representing the past           | Lecture/Exercises  |
| 3 Belief geometry in transformers | Reading/Discussion |

Questions?